Replication package README

Study and release metadata

- **Article:** *Density Forecasts and the Evolution of Macroeconomic Uncertainty in India*
- **Manuscript:** IJF-D-25-01079
- **Frozen release assembled:** October 8, 2026.
- **Github Repository:** https://github.com/karanbhasin95/rbi-spf-replication
- **Release license:** Creative Commons Attribution 4.0 International (`CC-BY-4.0`) for original material created by the authors; third-party material is excluded unless separately stated.
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This folder self-contained replication release. It contains the Stata dataset, the Stata code used by every computed table and figure, the final table and figure results, and the README files. Tables 1-5 and 7, Figures 1-8, and Appendix Figures D.1-D.2 all run from `data/baseline_data.dta`. Table 6 is descriptive and therefore has no do-file.

The frozen folder was independently run after assembly and generated all final table and figure files.

Quick start

Open PowerShell in this folder and run:

```
.\run_all.ps1
```

The launcher checks `data/baseline_data.dta`, locates StataNow 19 or Stata 19, runs `code/master.do`, and verifies a temporary completion marker that is deleted after the run. All paths are relative to this folder. The run writes the final tables to `results/tables` and the final figure panels to `results/figures`. Intermediate calculations remain internal to Stata and are not exported as repository results.

No network connection, credentials, GPU, raw-source download, or Python installation is required for the Stata replication. Expected runtime is approximately 2 minutes on the tested.

Folder structure

Location	Contents and role
`data/`	`baseline_data.dta`, the dataset.
`code/master.do`	Master Stata program that runs every computed table and figure.
`code/tables/`	Six dedicated Stata programs for Tables 1-5 and 7.
`code/figures/`	Ten self-contained Stata programs for Figures 1-8 and Appendix Figures D.1-D.2.
`results/tables/`	One final CSV for each manuscript table.
`results/figures/`	Final PNG panels only.

Computing environment

Component	Tested configuration	Requirement or role
Operating system	Microsoft Windows 11 Pro, 64-bit, build 26300	The supplied launcher is a PowerShell script for Windows.
Processor	AMD Ryzen AI MAX+ 395 with Radeon 8060S; 32 logical processors	No special CPU instruction set is required.
Memory	63.6 GB RAM	
PowerShell	7.6.5	Runs `run_all.ps1`.
Stata	StataNow MP 19.5, configured for 2 cores	Required for the complete analysis; Stata is proprietary and requires a valid license.
Stata add-ons	None	The analysis uses official Stata and Mata commands only.
GPU	Not used	No accelerator setup is needed.
Network	Not used	The frozen run makes no live downloads.

If Stata is installed in a different location, update the candidate executable paths near the top of `run_all.ps1`.

Runtime data

`data/baseline_data.dta` is the sole dataset read by the table and figure programs. Its `record_type` field separates the following observation groups:

- `annual_gdp`: 73 annual real-GDP level observations used by Table 1 and as realized outcomes elsewhere;
- `macro_history`: 74 retained annual GDP-growth and CPI-inflation observations used by Figure 1;
- `spf_histogram`: 1,676 RBI SPF histogram-bin observations used to derive Tables 2-5 and Figures 2-4;
- `monthly_inflation`: 201 realized monthly inflation observations used to construct Table 4 fiscal-year outcomes;
- `quarterly_macro`: 77 quarterly macroeconomic observations;
- `figure_04_infl_error`: 60 inflation forecast and actual pairs;
- `fitted_growth`: 62 retained fitted-density mean and uncertainty observations;
- `growth_density_fit`: 652 fitted nowcast beta-density bin observations used to derive unscaled skewness for the Appendix D.2 SVAR;
- figure09_density_fit: 753 fitted year-ahead beta-density bin observations used to calculate the Figure 9 / Appendix D.1 chart's skewness;
- `rbi_disagreement`: 21 bimonthly fitted variance and RBI disagreement observations used by Figure 5;
- `monthly_uncertainty`: 186 monthly India EPU and India VIX observations used by Figure 6;
- `sensitivity_histogram`: 323 raw histogram-bin observations used to estimate Table 7; and
- `svar_model_input`: 67 surviving quarterly level observations used to estimate Figures 7, 8, and D.2.

Figures 7, 8, and D.2 estimate their SVARs and generate IRFs during each run.

Code and results mapping

Manuscript item	Main Stata code	Generated result
Table 1	code/tables/table_01.do	results/tables/table01.csv
Table 2	code/tables/table_02.do	results/tables/table02.csv
Table 3	code/tables/table_03.do	results/tables/table03.csv
Table 4	code/tables/table_04.do	results/tables/table04.csv
Table 5	code/tables/table_05.do	results/tables/table05.csv
Table 6	No do-file; descriptive manuscript table	Not applicable
Table 7	code/tables/table_07.do	results/tables/table07.csv
Figure 1	code/figures/figure_01.do	results/figures/figure01.png
Figure 2 panels a-d	code/figures/figure_02.do	results/figures/figure02a.png-figure02d.png
Figure 3 panels a-d	code/figures/figure_03.do	results/figures/figure03a.png-figure03d.png
Figure 4	code/figures/figure_04.do	results/figures/figure04.png
Figure 5	code/figures/figure_05.do	results/figures/figure05.png
Figure 6	code/figures/figure_06.do	results/figures/figure06.png
Figure 7 panels a-e	code/figures/figure_07.do	results/figures/figure07a.png-figure07e.png
Figure 8 panels a-e	code/figures/figure_08.do	results/figures/figure08a.png-figure08e.png
Figure 9 / Appendix Figure D.1	code/figures/appendix_figure_d1.do	results/figures/appendix_figure_d1.png
Appendix Figure D.2 panels a-e	code/figures/appendix_figure_d2.do	results/figures/appendix_figure_d2a.png-appendix_figure_d2e.png

Results

The results contain seven table CSVs and 28 PNG figure panels.

Licensing

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